

House Price Prediction – Project

Summary

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Overview

The objective of this project was to develop a machine learning model capable of predicting house prices using various property features such as area, number of bedrooms, bathrooms, parking availability, furnishing status, and other amenities. The dataset used for this analysis was the Housing Prices Dataset obtained from Kaggle. Data preprocessing was performed to handle categorical variables and prepare the dataset for machine learning algorithms.

Methodology

The dataset was explored to identify missing values, duplicates, and feature relationships. Categorical attributes were converted into numerical format using one-hot encoding. The cleaned dataset was then divided into training and testing sets using an 80:20 ratio. Two regression models were trained and evaluated:

- Linear Regression
- Random Forest Regressor

Model performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Results

The Linear Regression model achieved the best performance with an R^2 score of **65.29%**, MAE of **970,043**, and RMSE of **1,324,507**. The Random Forest Regressor achieved an R^2 score of **61.33%**, MAE of **1,013,969**, and RMSE of **1,398,116**.

Feature importance analysis revealed that **area** was the most influential factor in determining house prices, followed by **bathrooms**, **air conditioning**, **parking**, and **number of stories**. Correlation analysis also confirmed that larger properties with better amenities generally have higher market values.

Key Insights

- Property area is the strongest predictor of house price.
- Bathrooms, parking spaces, and air conditioning significantly contribute to property value.
- Most houses in the dataset fall within the mid-price range, while luxury properties are relatively uncommon.
- Linear Regression outperformed Random Forest on this dataset, indicating that the relationship between features and price is reasonably linear.

Conclusion

The project successfully demonstrated how machine learning can be used to estimate house prices based on property characteristics. The findings highlight the importance of property size and amenities in determining market value. Such predictive models can assist real estate businesses, buyers, and sellers in making more informed pricing decisions and improving valuation accuracy.